

Tracking & Measurement Plan

GA4 + GTM Governance Framework and Event Taxonomy — Coursera Practice Build

PERFORMANCE
MARKETING
ANALYTICS

Document owner: Analytics/Tracking Lead

Applies to: GTM container `GTM-T8H79C7H` and any future container built on the same standard

Audience: Leadership (scope, coverage, gaps), engineering/analytics (naming, implementation, QA)

This document defines how tracking is designed, named, built, tested, documented, and maintained across our GA4/GTM implementations, and records the specific event taxonomy currently built for the Coursera practice container.

Part A is the governance framework: container structure, naming rules, QA process, documentation standard, access control, privacy rules, and audit cadence. It applies to any property, not just this one.

Part B is the applied result: the concrete event taxonomy built for the Coursera practice site — what's live, what it measures, and what's still open — serving as the template for the next real-world rollout.

Table of Contents

Part A — Tracking Governance & Management Plan

- A.1 Container Architecture
- A.2 Naming Convention
- A.3 Data Layer Specification
- A.4 QA & Publishing Workflow
- A.5 Event Dictionary
- A.6 Access & Change Management
- A.7 Privacy & Consent
- A.8 Audit Cadence
- A.9 Known Issues & Recommendations

Part B — Coursera Practice Build: Event Taxonomy & Status

- B.1 Baseline Layer
- B.2 Funnel Layer
- B.3 Engagement & Content Layer
- B.4 Event-to-Goal Mapping
- B.5 Roadmap / Next Steps

Part A — Tracking Governance & Management Plan

A.1 Container Architecture

- **One GTM container per property.** Web and app containers are never mixed.
- **Workspace discipline:** all work happens in a named workspace (`WIP - [name] - [feature]`); the shared default workspace is never edited directly once a team exists.
- **Version discipline:** every publish creates a GTM Version automatically — versions are never deleted; this is the rollback safety net. Each version is named with what changed.
- **Folders**, organized by function, not by date:
 - `01_Config` — GA4 Configuration tag, constants, global built-ins
 - `02_Ecommerce_Events` — the GA4 standard funnel schema
 - `03_Engagement_Events` — auth, forms, header/footer clicks, signup timing
 - `04_UX_Technical_Events` — the GA4-facing half of custom UX/technical-health signals
 - `05_Custom_Listeners` — the Custom HTML "engine" tags that generate the dataLayer pushes consumed by `04`, plus the DOM_READY triggers gating the ecommerce listener tags
 - `99_Deprecated` — retired tags moved here before deletion, never deleted same-day

Current Container Inventory

`01_Config`

Type	Name
Tag	GA4 - config
Trigger	Window Loaded
Variable	Constant - measurement_id , DLV - page_path

02_Ecommerce_Events

Type	Name
Tag	GA4E - view_item , GA4E - select_item , GA4E - begin_checkout , GA4E - add_to_cart , GA4E - remove_from_cart , GA4E - spotlight_collections_1
Trigger	CE - view_item , CE - select_item , CE - begin_checkout , Click - add_to_cart , Click - remove_from_cart , CE - spotlight_collections_1
Variable	DLV - interacted_form_name

03_Engagement_Events

Type	Name
Tag	GA4E - login , GA4E - logout , GA4E - form_submit , GA4E - social_footer , GA4E - header_explore , GA4E - header_segments , GA4E - footer_clicks , GA4E - and_ios_button , GA4E - signup , GA4E - high_intent_visitors , GA4E - scroll_tracking , CHTML - set_entry_time , CHTML - form_abandonment
Trigger	CE - login , CE - logout , CE - form_submit , Click - social_footer , Click - header_explore , Click - any_button , Click - and_button , Click - ios_button , Click - header_segmentation , Click - footer_elements , CE - signup , Scroll Depth - all_pages , Scroll Depth - 75_percent , Timer - 30_sec , Trigger Group - high_intent_visitors , Initialization - set_entry_time , DOM - copy_listener_init , DOM - form_abandonment
Variable	CJS - time_spent_before_signup , DLV - copied_text_preview , DLV - form_id , DLV - field_name , DLV - field_has_value , DLV - last_field_touched , DLV - scroll_depth_percent , DLV - time_on_page_sec

04_UX_Technical_Events

Type	Name
Tag	GA4E - rage_click , GA4E - text_copied , GA4E - form_abandonment , GA4E - exit_intent , GA4E - dead_click , GA4E - adblock_detection , GA4E - checkout_tab_switch , GA4E - file_print_or_save , GA4E - network_quality_measure , GA4E - text_highlight , CHTML - dead_click_listener_init
Trigger	CE - rage_click , CE - text_copied , CE - form_abandonment , CE - exit_intent , CE - dead_click , CE - adblock_status , CE - checkout_tab_switch , CE - network_quality , CE - print_page_action , CE - text_selection , DOM - rage_click_listener_init , DOM - dead_click_listener_init
Variable	DLV - copied_text_length , DLV - element_text , DLV - element_class , DLV - element_id , DLV - adblocker_active , DLV - tab_action , DLV - exit_trigger_type , DLV - network_downlink , DLV - network_rtt , DLV - network_type , DLV - selected_text , DLV - selected_text_length , DLV - time_away_sec

05_Custom_Listeners

Type	Name
Tag	CHTML - rage_click_listener , CHTML - text_copy_listener , CHTML - exit_intent_listener , CHTML - adblocker_detection , CHTML - checkout_tab_switch_tracker , CHTML - network_quality_tracker , CHTML - print_action_tracker , CHTML - text_selection_tracker , CHTML - spotlight_collections_1 , CHTML - select_item , CHTML - view_item , CHTML - begin_checkout
Trigger	DOM - spotlight_collections_1 , DOM - select_item , DOM - view_item , DOM - begin_checkout
Variable	DLV - element_tag

Unfoldered

Type	Name	Note
Trigger	<code>Click - just_links</code> , <code>Click - all_elements</code>	Generic triggers, not currently used by any tag

A.2 Naming Convention

Element	Pattern	Example
GA4 Configuration tag	<code>GA4 - {purpose}</code>	<code>GA4 - config</code> — the only tag allowed to keep the bare <code>GA4</code> prefix
GA4 Event tag	<code>GA4E - {event_name}</code>	<code>GA4E - rage_click</code>
Custom HTML tag	<code>CHTML - {description}</code>	<code>CHTML - text_copy_listener</code>
Custom Event trigger	<code>CE - {event_name}</code>	<code>CE - exit_intent</code>
Click trigger	<code>Click - {description}</code>	<code>Click - social_footer</code>
DOM Ready trigger	<code>DOM - {description}</code>	<code>DOM - rage_click_listener_init</code>
Scroll Depth trigger	<code>Scroll Depth - {description}</code>	<code>Scroll Depth - 75_percent</code>
Timer trigger	<code>Timer - {interval}</code>	<code>Timer - 30_sec</code>
Trigger Group	<code>Trigger Group - {description}</code>	<code>Trigger Group - high_intent_visitors</code>
Initialization trigger	<code>Initialization - {description}</code>	<code>Initialization - set_entry_time</code>
Built-in trigger	unprefixed, GTM default name	<code>Window Loaded</code>
Data Layer Variable	<code>DLV - {key}</code>	<code>DLV - element_tag</code>
Custom JS Variable	<code>CJS - {description}</code>	<code>CJS - time_spent_before_signup</code>
Constant Variable	<code>Constant - {description}</code>	<code>Constant - measurement_id</code>
dataLayer event name	<code>snake_case</code> , verb_noun	<code>rage_click</code>

Rules:

- Descriptive/event-name portions are strict `snake_case` — no Title Case, no spaces, no stray hyphens inside a name segment.
- Exactly one space on each side of the separator: `{Prefix} - {name}`.
- `GA4` is reserved for the single base configuration tag; every event-sending tag is `GA4E`.
- Click-type triggers use a single unified `Click` prefix.
- The tag name should tell you what fires and why, without opening it.

A.3 Data Layer Specification

Every custom `dataLayer.push()` is documented before it's built — treated as an API contract:

```
Event: {event_name}
Fires when: {plain-language trigger description}
Internal dataLayer event (only if different from the GA4 event name): {value}
Parameters:
  - {key}: {type} — {what it means}
Business question it answers: {one sentence}
Owner: {who added it, when}
Status: Active / Deprecated
```

The "internal dataLayer event" line matters because the dataLayer `event` key that fires a GTM trigger and the `eventName` parameter actually sent to GA4 are not required to match — several events in this build use that pattern intentionally (see A.9). Recording both prevents confusion in DebugView.

A.4 QA & Publishing Workflow

1. Build in workspace → GTM Preview mode against a real session → confirm tag fires on correct trigger, correct parameters

- 2. Cross-check in GA4 DebugView — the event must appear with correct parameter values, not just "fired"
- 3. Run the Pre-Publish Checklist
- 4. Publish with a version name describing the change
- 5. Spot-check in GA4 Realtime within 10 minutes of publish

PRE-PUBLISH CHECKLIST

- ☐ No tag fires on every page unintentionally
- ☐ No duplicate events firing
- ☐ Parameter values are real data, not `undefined` / `null`
- ☐ Event name matches the Data Layer Spec exactly, including the internal-event vs. GA4-event distinction where it applies
- ☐ No PII in any parameter

A.5 Event Dictionary

The single source of truth for what's live. `Internal event` is listed only where it differs from the GA4 event name.

GA4 event	Internal event	Category	Folder	Fires when	Key parameters	Business question	Status
<code>config</code> (base)	—	Config	01	Page load	<code>measurement_id</code>	Base GA4 pageview stream	Active
<code>scroll</code>	—	Baseline	03	10/25/50/75/90% scroll	<code>percent_scrolled</code> , <code>page_location</code>	Are people reading past the fold?	Active
<code>view_item_list</code>	—	Funnel	02	Spotlight collection section visible 3s	<code>item_list_name</code> , <code>items[]</code>	What catalog content gets seen?	Active
<code>select_item</code>	—	Funnel	02	Click a course card	<code>item_id</code> , <code>item_name</code> , <code>item_brand</code> , <code>index</code>	Which cards convert clicks?	Active
<code>view_item</code>	—	Funnel	02	Course detail page load	<code>item_id</code> , <code>item_name</code> , <code>item_brand</code>	Course-page reach	Active
<code>add_to_cart</code>	—	Funnel	02	Click "Start Free Trial" button	(none configured)	Free-trial intent volume	Active — needs UI review, see A.9
<code>begin_checkout</code>	—	Funnel	02	Checkout page reached	<code>currency</code> , <code>value</code> , <code>items[]</code>	Checkout entries	Active
<code>remove_from_cart</code>	—	Funnel	02	Click text "Remove from cart"	(none configured)	Cart abandonment signal	Active — needs UI review, see A.9
<code>login</code>	(built-in <code>gtm.formInteract</code>)	Engagement	03	Interaction with <code>.rc-LoginForm</code>	(none configured)	Login activity	Active
<code>logout</code>	(built-in <code>gtm.formSubmit</code>)	Engagement	03	Submit on a <code>LogoutSsr</code> form	(none configured)	Logout activity	Active
<code>generate_lead</code>	(pageview)	Engagement	03	Pageview on <code>/thank_you_plus</code>	<code>form_id</code> (built-in)	Lead-form completions	Active
<code>click</code>	—	Engagement	03	Click inside header segmentation nav	<code>link_classes</code> , <code>link_url</code> , <code>link_text</code>	Header nav usage by segment	Active
<code>footer_click_elements</code>	—	Engagement	03	Click a footer link	(none)	Footer engagement	Active
<code>footer_apk_download</code>	—	Engagement	03	Click Android/iOS app badge	<code>link_url</code>	App-download intent	Active
<code>social_footer_button</code>	—	Engagement	03	Click a footer social icon	<code>link_url</code>	Social referral intent	Active
<code>header_explore</code>	—	Engagement	03	Click a link inside the mega-menu	<code>link_url</code> , <code>link_text</code>	Explore-menu usage	Active
<code>sign_up</code>	—	Engagement	03	Pageview with <code>isNewUser=true</code>	<code>time_to_signup_secs</code>	Signup volume + time-to-decide	Active
<code>high_intent_visit or</code>	—	Engagement	03	30s on page and 75% scroll (Trigger Group)	<code>intent_level</code> , <code>trigger_criteria</code> , <code>page_type</code>	Warm-lead identification	Active

GA4 event	Internal event	Category	Folder	Fires when	Key parameters	Business question	Status
<code>text_copied</code>	—	Content value	04	<code>copy</code> event with a selection	<code>copied_length</code> , <code>copied_preview</code> , <code>page_path</code>	What content gets saved/copied?	Active
<code>rage_click</code>	—	UX health	04	3+ clicks on same element within 1s	<code>element_tag</code> , <code>element_text</code> , <code>element_class</code> , <code>page_path</code>	Where is the UI frustrating?	Active
<code>dead_click</code>	—	UX health	04	Click on a non-interactive element	<code>element_tag</code> , <code>element_text</code> , <code>element_class</code> , <code>element_id</code> , <code>page_path</code>	Where do users expect something to happen and nothing does?	Active
<code>form_abandon</code>	<code>form_abandonment</code>	Funnel friction	04	Tab hidden after a touched, unsubmitted field	<code>form_id</code> , <code>last_field</code>	Exact drop-off field	Active
<code>exit_intent</code>	—	True engagement	04	Mouse leaves top of viewport, or tab hidden after 5s	<code>trigger_type</code> , <code>time_on_page</code> , <code>scroll_depth</code>	Who's about to leave, and how far did they get?	Active
<code>adblock_detection</code>	<code>adblock_status</code>	UX health	04	Bait-element measurement, 300ms after load	<code>adblocker_active</code>	Ad-blocker prevalence (data-completeness signal)	Active
<code>checkout_tab_switch</code>	—	Funnel friction	04	Returns to a checkout tab after ≥1s away	<code>tab_action</code> , <code>time_away_seconds</code>	Checkout hesitation / comparison shopping	Active
<code>file_print_or_save</code>	<code>print_page_action</code>	Content value	04	Browser print dialog (<code>beforeprint</code>)	<code>page_location</code>	Which pages get printed/saved?	Active
<code>network_quality_measure</code>	<code>network_quality</code>	Technical	04	Page has a <code>navigator.connection</code> API	<code>network_type</code> , <code>rtt</code> , <code>download</code>	Segment behavior by connection quality	Active
<code>text_highlight</code>	<code>text_selection</code>	Content value	04	Text selection held ≥800ms, ≥10 chars	<code>selected_text</code> , <code>text_length</code>	What passages get highlighted for attention/study?	Active

A.6 Access & Change Management

- GTM user roles: **Publish** (senior review stage only), **Edit** (working level), **View** (stakeholders who need visibility without edit rights)
- Every publish gets a one-line change log entry: what changed, why, expected impact
- No untested change is made directly to the live container mid-campaign — the same discipline as never changing bidding strategy and reallocating budget in the same week

A.7 Privacy & Consent

- Tags respect Consent Mode v2 state rather than firing unconditionally
- No PII in any parameter — no emails, names, phone numbers, or free-text form contents
- `copied_preview` (from copy tracking) sends up to 50 characters of copied text, and `text_highlight`'s `selected_text` sends up to 100 characters of a held selection. Both need periodic review against the no-PII rule, since a user could copy or highlight their own name, email, or other personal text on a form or profile page — either scope these listeners away from form/profile pages, or add a lightweight PII pattern check before the push.
- Practice-site rule: all injected tracking on the Coursera practice site runs client-side, in-browser only, via the extension — nothing is published to Coursera's real infrastructure, no real forms are submitted, no real purchases are completed, and no unusual request volume is sent to their servers.

A.8 Audit Cadence

- **Monthly**: review for duplicate/redundant tags and unused triggers
- **Quarterly**: full container audit against the Event Dictionary — anything undocumented gets documented or removed

A.9 Known Issues & Recommendations

- `CE - form_submit` is a `PAGEVIEW`-type trigger (fires on `/thank_you_plus`), not a true custom-event trigger — the `CE -` prefix is misleading here. Recommend renaming to `Page View - form_submit`, or converting it to a genuine `generate_lead` dataLayer push if a reliable dataLayer signal is available.
- `CE - signup` is a `DOM_READY`-type trigger (fires when the URL contains `isNewUser=true`), same mismatch — recommend `DOM - signup`.

- Several tags use an internal dataLayer event name that differs from the GA4 event name they send (`adblock_status` → `adblock_detection` , `network_quality` → `network_quality_measure` , `print_page_action` → `file_print_or_save` , `text_selection` → `text_highlight` , `form_abandonment` → `form_abandon`). This is a valid pattern, but every one of these must carry the "internal event" line in its Data Layer Spec (A.3) or it costs real time in DebugView.
- **GA4E - add_to_cart** fires on a "Start Free Trial" button click and **GA4E - remove_from_cart** fires on link text "Remove from cart" — confirm both still map to real, current Coursera UI elements; they read as carried over from a generic ecommerce template rather than purpose-built for a course-enrollment flow.
- **CHTML - form_abandonment** (the listener) sits in `03_Engagement_Events` , while its GA4 tag and trigger sit in `04_UX_Technical_Events` . Inconsistent with the "listener engines in 05, GA4-facing output in 04" pattern used elsewhere — move the listener to `05_Custom_Listeners` , or move the pair back to `03` .
- **DLV - interacted_form_name** sits in `02_Ecommerce_Events` but reads a form-interaction built-in (`gtm.interactedFormName`) and appears unused by any tag — re-home to `03_Engagement_Events` or move to `99_Deprecated` if confirmed dead.
- **Click - just_links** and **Click - all_elements** are unfoldered and currently fire no tags — assign to a folder with a documented purpose, or move to `99_Deprecated` .

Part B — Coursera Practice Build: Event Taxonomy & Status

B.1 Baseline Layer

Event	Status	Notes
<code>scroll_depth</code>	● Live	10/25/50/75/90% via <code>Scroll Depth - all_pages</code> ; a dedicated <code>75_percent</code> trigger also feeds <code>high_intent_visitor</code>
<code>outbound_click</code>	○ Planned	No tag/trigger pair configured yet
<code>file_download</code>	○ Planned	—
<code>site_search</code>	○ Planned	—
<code>video_engagement</code>	○ Planned	See B.3 #10

B.2 Funnel Layer

GA4 event	Status	Notes
<code>view_item_list</code>	● Live	Built for the <code>spotlight_collections_1</code> module specifically; not yet generalized to other list surfaces (e.g. search results)
<code>select_item</code>	● Live	
<code>view_item</code>	● Live	
<code>add_to_wishlist</code>	○ Planned	—
<code>begin_checkout</code>	● Live	
<code>add_payment_info</code>	○ Planned	—
<code>purchase</code>	○ Planned	Largest remaining funnel gap — no confirmed-enrollment event exists yet
<code>add_to_cart</code> / <code>remove_from_cart</code>	● Live, needs review	See A.9

B.3 Engagement & Content Layer

#	Item	Status	Notes
1	Copy-to-clipboard tracking	● Live	<code>text_copied</code>
2	Rage-click detection	● Live	3+ clicks on same element within 1s
3	Dead-click detection	● Live	Also captures <code>element_id</code>
4	Form field abandonment	● Live	Uses <code>focusout</code> delegation + <code>visibilitychange</code> , pushed directly to <code>dataLayer</code> ; reports <code>last_field_touched</code>
5	Active engagement time	○ Planned	No <code>engaged_time</code> event yet
6	Element visibility / impression tracking	● Partial	The <code>IntersectionObserver</code> pattern is used for <code>view_item_list</code> ; a generic "any element visible" version isn't built
7	SPA virtual pageviews	○ Planned	Coursera's course-player route changes aren't currently visible to analytics
8	JS error tracking	○ Planned	—
9	Exit-intent detection	● Live, extended	Combines <code>mouseleave</code> (top of viewport) with a 5s-minimum tab-switch signal; reports <code>time_on_page</code> + <code>scroll_depth</code>
10	Video milestone tracking	○ Planned	—

Additional events live beyond the original taxonomy:

Event	What it tracks	Business question
<code>network_quality_measure</code>	Connection type, RTT, downlink via <code>navigator.connection</code>	Does site/video performance correlate with connection quality?
<code>adblock_detection</code>	Bait-element measurement	How much of the audience runs an ad blocker (data-completeness caveat)?
<code>file_print_or_save</code>	<code>beforeprint</code> event	Which pages get printed or saved as PDF?
<code>text_highlight</code>	Held text selection $\geq 800\text{ms}$, ≥ 10 chars	What content is worth highlighting/studying?
<code>checkout_tab_switch</code>	Tab hidden/returned while on <code>/checkout</code>	Comparison-shopping or hesitation during checkout

B.4 Event-to-Goal Mapping

Category	Events	Business question it answers
Content value	<code>text_copied</code> , <code>text_highlight</code> , <code>file_print_or_save</code> , <code>view_item_list</code> / <code>view_item</code>	What content actually gets consumed, saved, or valued?
UX health	<code>rage_click</code> , <code>dead_click</code> , <code>adblock_detection</code>	Where is the interface frustrating, broken, or under-measured?
Funnel friction	<code>form_abandon</code> , <code>checkout_tab_switch</code> , <code>begin_checkout</code> without <code>purchase</code>	Exactly which step loses people
True engagement	<code>exit_intent</code> , <code>high_intent_visitor</code>	Real attention and intent, not inflated time-on-page
Technical health	<code>network_quality_measure</code>	Whether performance is a hidden driver of drop-off
Navigation accuracy	(gap — <code>virtual_pageview</code> not yet built)	Whether analytics sees the whole site, including SPA sections

B.5 Roadmap / Next Steps

1. Close the funnel gap: `purchase`, `add_to_wishlist`, `add_payment_info`
2. Fix the biggest measurement blind spot: **SPA virtual pageviews** for the course player
3. Resolve the A.9 naming/folder items before the next quarterly audit
4. Keep the Event Dictionary (A.5) as the living source of truth going forward
5. Once `purchase` and virtual pageviews are live, this taxonomy structure is ready to be re-applied as the real production Phase 4 setup